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(54) **GEOMETRIC CONCEPT FOR A
ROLLER-FLANGE CONTACT IN ROLLER
BEARINGS**

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(2013.01); **F16C 19/225** (2013.01); **F16C**
2240/40 (2013.01); **F16C 19/364** (2013.01)

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F16C 33/585; F16C 33/586; F16C 33/64;
F16C 2240/30; F16C 2240/40; F16C 2240/50
USPC 384/564, 565, 571
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

6,530,693 B1 3/2003 Ijuin
8,033,737 B2 * 10/2011 De Mul 384/564
2007/0041678 A1 2/2007 Matsuyama

FOREIGN PATENT DOCUMENTS

CH 212693 A 12/1940
DE 10203113 A1 8/2002

(Continued)

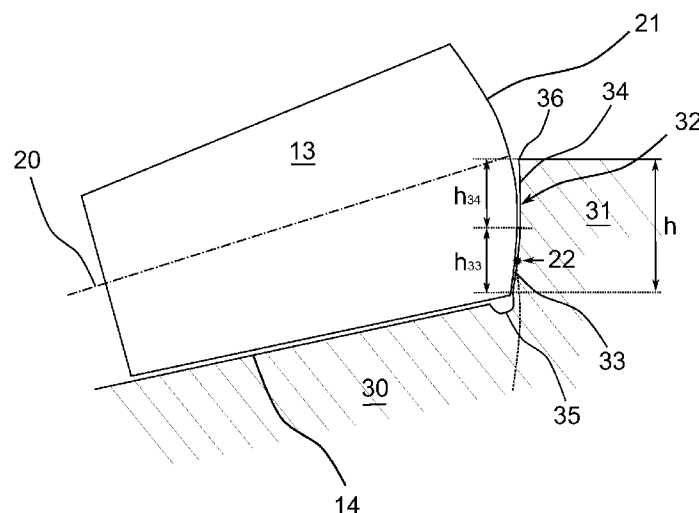
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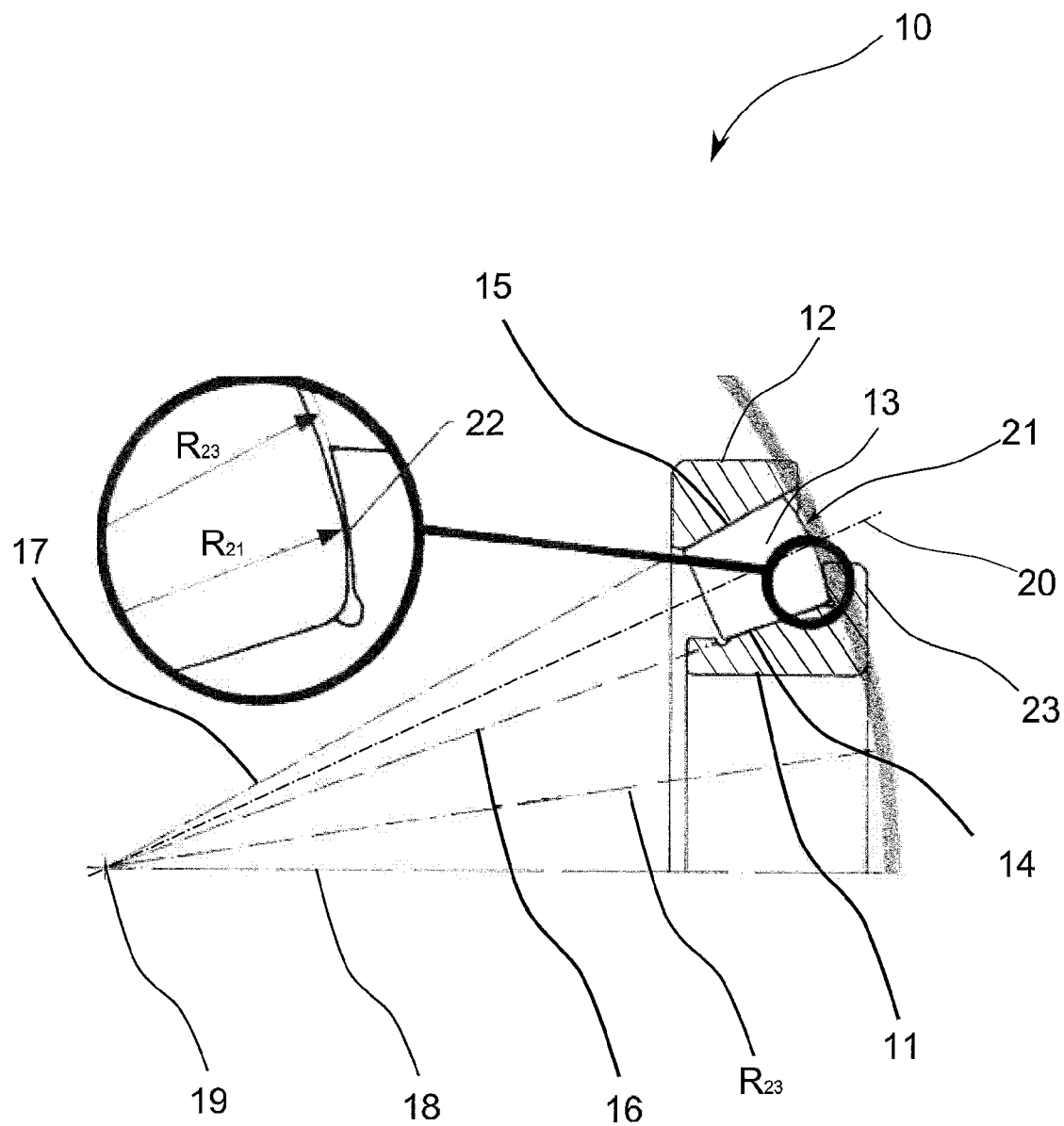
(57) **ABSTRACT**

A bearing ring for a roller bearing includes a raceway for at least one roller that includes an at least partially spherically formed roller end surface having a first curvature. A flange for transferring axial forces extends from an end of the raceway and has a flange surface that faces towards the roller end surface. A spherical flange surface section having a second curvature is defined on the flange surface. The first curvature differs from the second curvature such that a first gap (d_1) exists between the spherically formed roller end surface and the spherical flange surface section. The flange surface also includes an opening flange surface section having a third curvature, which is less than the second curvature, such that a second gap (d_2) exists between the roller end side and the opening flange surface section. The second gap is larger than the first gap.

20 Claims, 6 Drawing Sheets



(56)	References Cited			
		EP	1482191 A2	12/2004
		EP	1754899 A2	2/2007
		JP	4331813 A	5/2000
	FOREIGN PATENT DOCUMENTS	JP	2005003121 A	1/2005
		JP	2006349014 A	12/2006
DE	102005061102 A1			7/2007
DE	102008020068 A1			10/2009
		* cited by examiner		



PRIOR ART

Fig. 1

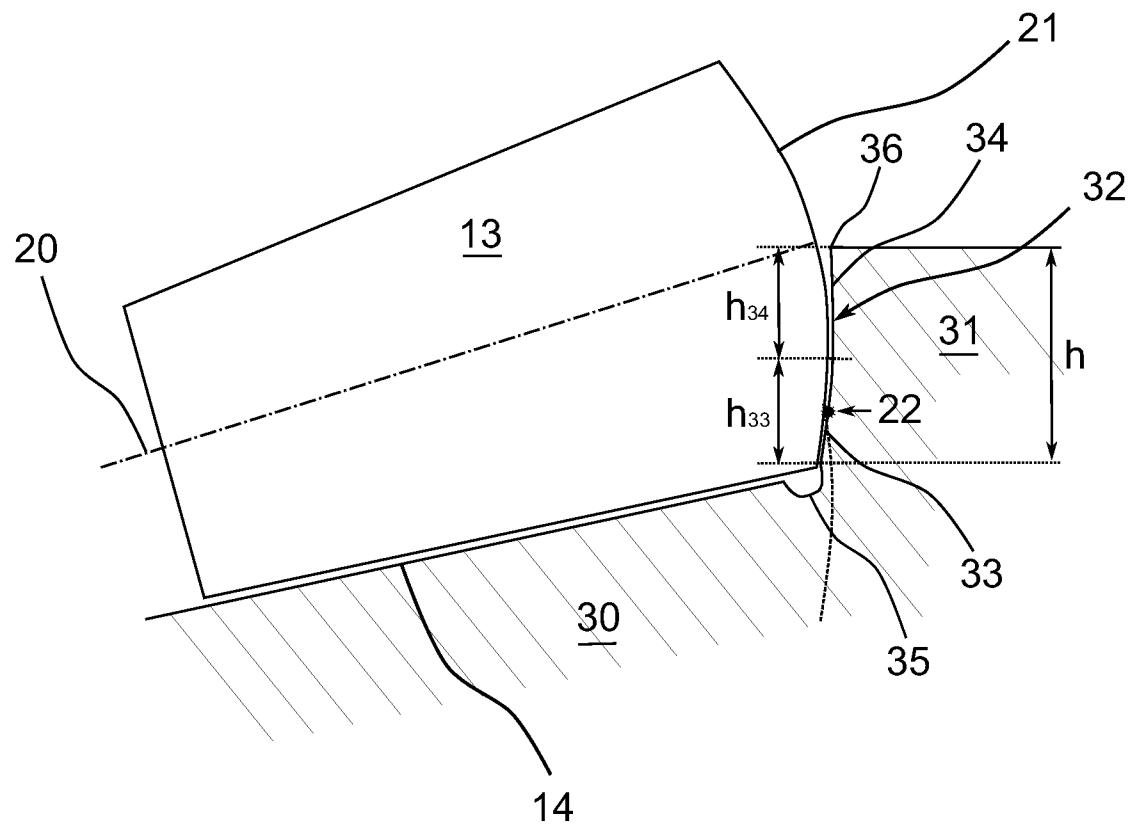


Fig. 2

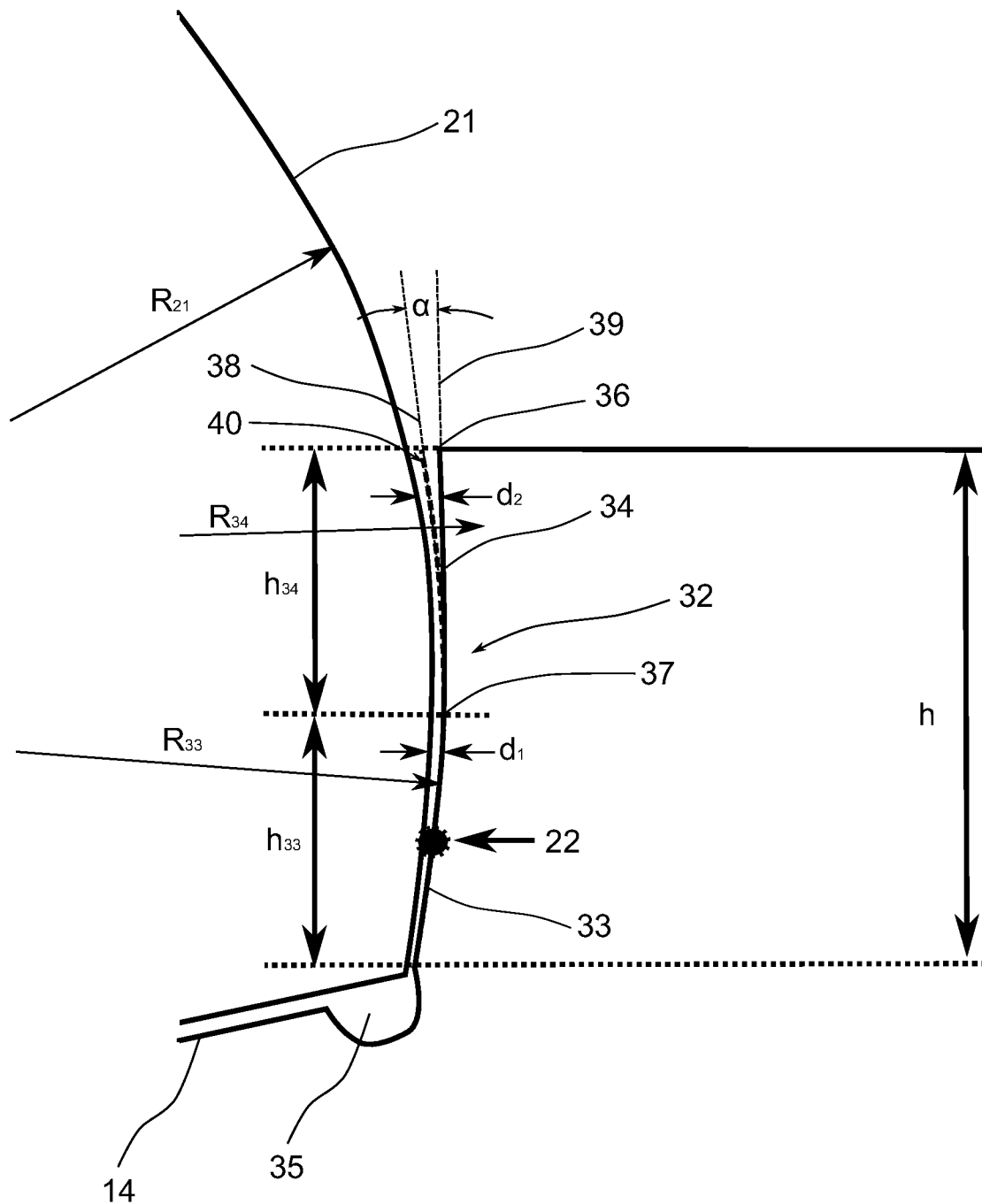


Fig. 3

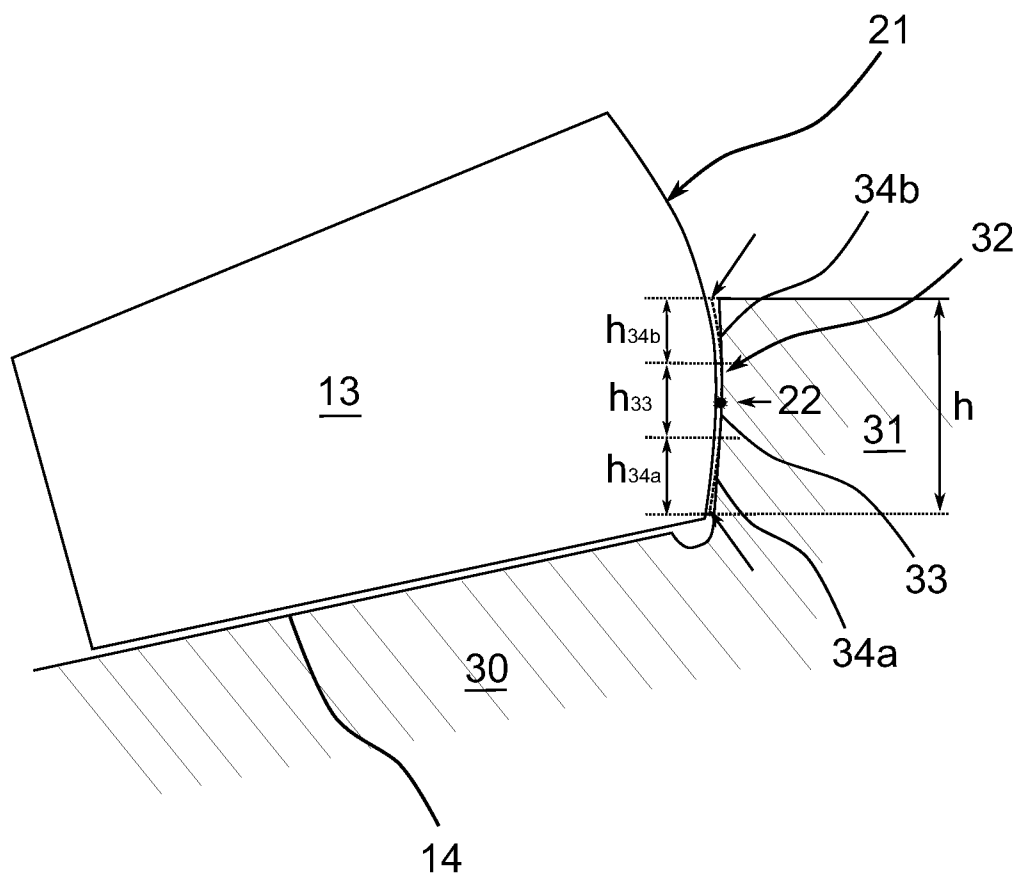


Fig. 4

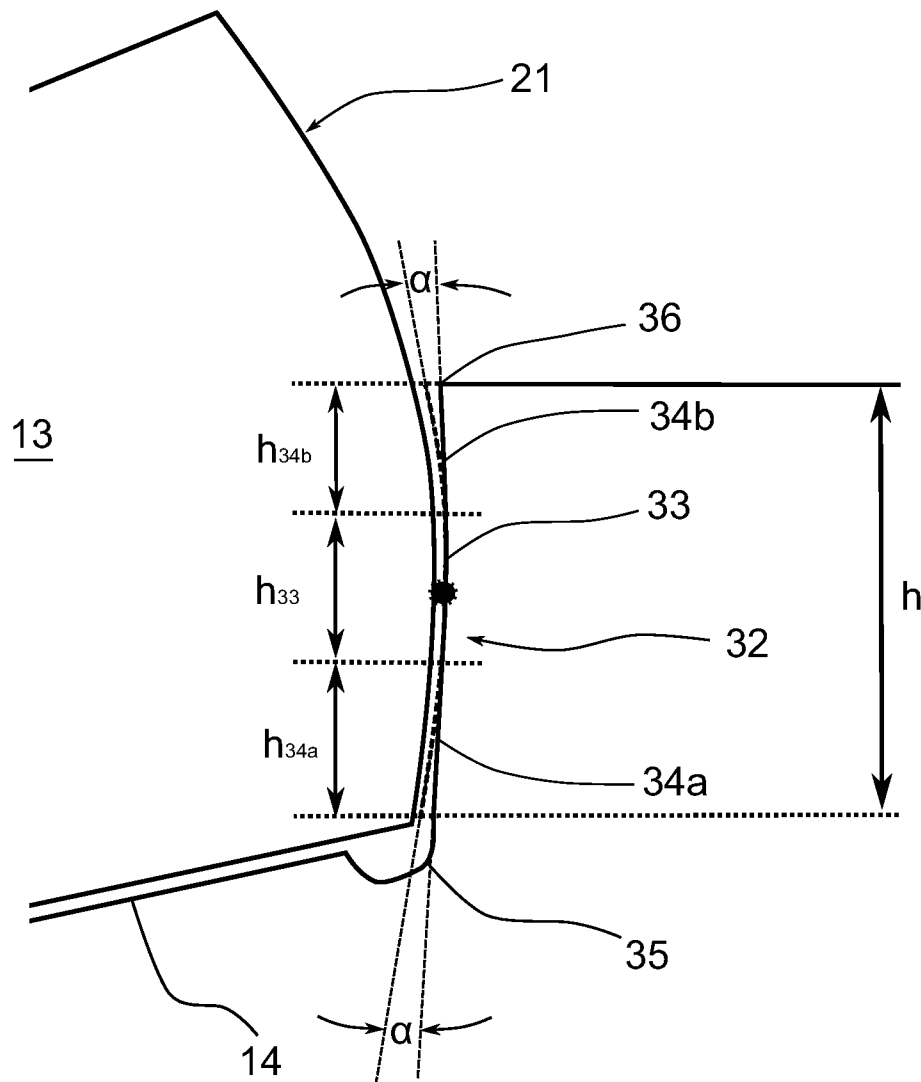


Fig. 5

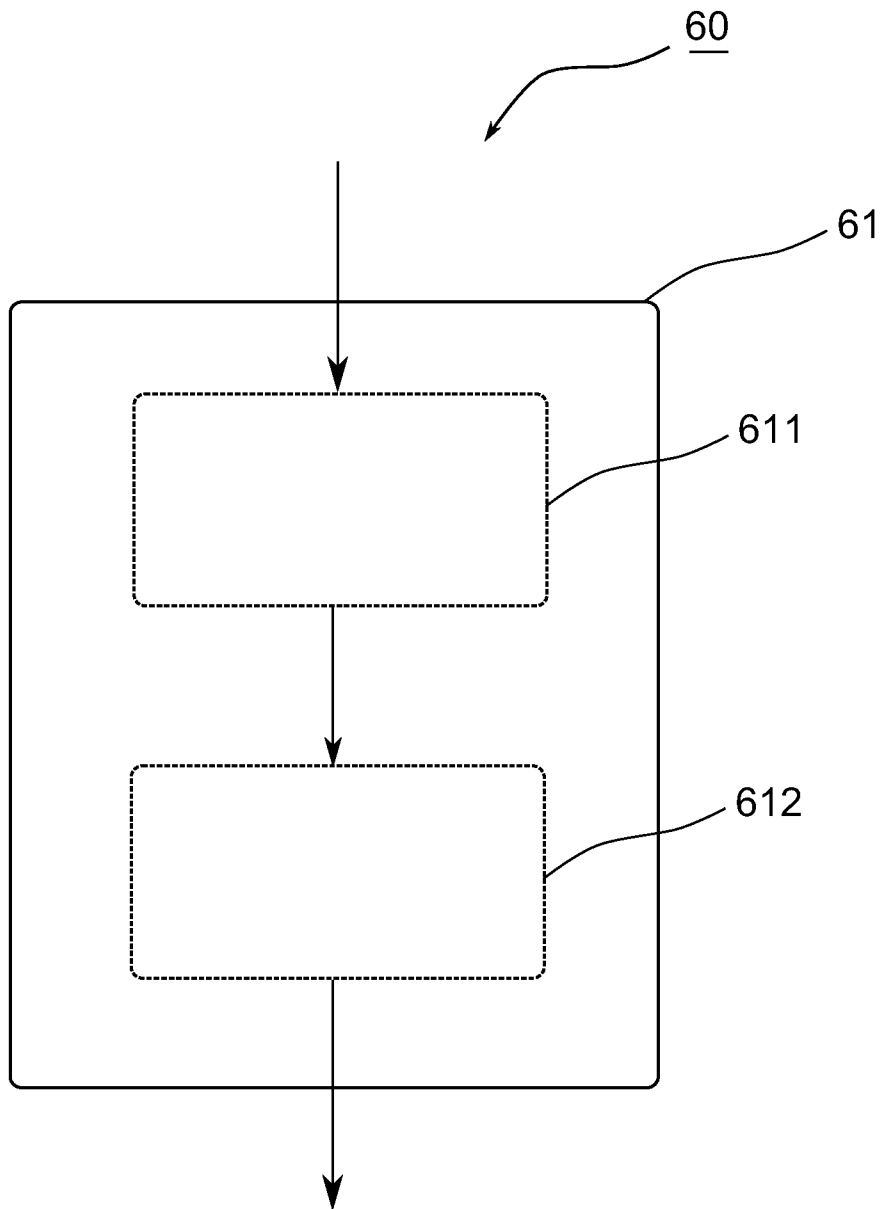


Fig. 6

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GEOMETRIC CONCEPT FOR A ROLLER-FLANGE CONTACT IN ROLLER BEARINGS

CROSS-REFERENCE

This application is the U.S. national stage of International Application No. PCT/EP2011/071157 filed on Nov. 28, 2011, which claims priority to German patent application no. 10 2010 062 481.0 filed on Dec. 6, 2010.

TECHNICAL FIELD

The present invention relates to a geometric concept for a roller-flange contact in roller bearings, in particular for a tapered-roller-flange-contact in tapered roller bearings.

RELATED ART

Guide flanges in roller bearings, such as e.g. tapered roller bearings, can be embodied either straight or spherical. Straight flanges are mainly used for roller bearings of small diameter and thus also of smaller flange widths. In this case, a flange width is often too small to produce a defined profile on an available flange surface. Spherical flanges find application mostly in roller bearings having larger diameters and thus also having larger flange widths. A spherical flange is characterized in that a radius, which describes a flange shape facing towards a roller raceway, has its origin substantially on an axis of rotation of the rollers (roller rotational axis), wherein small deviations are allowed due to alignment errors.

For a more detailed description of roller bearings having spherical flanges, FIG. 1 shows, in a schematic representation, a longitudinal section of a roller bearing 10, which is exemplarily formed as a tapered roller bearing. The roller bearing 10 includes a bearing inner ring 11, a bearing outer ring 12, and a plurality of rollers 13, which can roll on races or raceways 14, 15 formed by the inner sides of the bearing rings 11, 12. In the case of tapered roller bearings, tapered rollers are the rollers corresponding to the rollers or rolling elements 13.

The tapered rollers 13 can roll on an inner raceway 14, which is formed in the bearing inner ring 11, and on an outer raceway 15, which is formed in the bearing outer ring 12. Considering a tapered roller bearing in three-dimensions, the raceways 14, 15 are formed as conical outer surfaces. In the longitudinal section shown in FIG. 1 of the tapered roller bearing 10, the raceways 14, 15 define, in an imaginary extension, an inner line 16 and an outer line 17, which meet on an axis of rotation 18 of the roller bearing 10 at a rotation point 19.

During an operation of the bearing 10, each (tapered) roller 13 rotates around its own roller axis 20, wherein an imaginary extension of the roller axis 20 also intersects with the pivot point 19. A rolling condition on the raceways 14, 15 for the tapered rollers 13 is realized by a relative position of inner line 16, outer line 17, bearing axis of rotation 18, and roller axis 20, which all intersect in the pivot point 19, so that with a relative rotation of bearing inner ring 11 and bearing outer ring 12, the tapered rollers 13 roll on the raceways 14, 15 substantially without slippage, and an amount of friction related thereto is minimized.

When supporting axial forces, in order to also optimize the friction that occurs in the axial direction i.e. in the direction of the bearing rotational axis 18, the rollers 13 used in roller bearings can have a first curvature, on their end side 21, identified by a first radius R_{21} , so that a part of a ball surface

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results as an end side surface of the rollers 13, which, as is indicated in the enlargement of FIG. 1, is in contact at a contact point 22 with a straight- or spherically-embodied flange 23, for example of the bearing inner ring 11. In order to achieve a defined contact point 22, the spherically curved flange surface has a second curvature defined by a second radius R_{23} , which curvature is less than the first curvature of the spherical end side surface 21 of the rollers 13.

In roller bearings, such as for example cylindrical roller, barrel roller, or ball bearings, which are designed with straight or flat flanges, in comparison to spherically embodied flanges the roller-flange contact has a higher surface pressure (Hertzian pressure) between the roller end side 21 and the flange surface facing towards this. Here the Hertzian pressure is understood to be the greatest pressure that prevails in the middle of the contact surface of two elastic bodies. If, such as with roller bearings with straight flanges, two elastic bodies (curved roller end side and straight or flat flange) are pressed against each other, then in the ideal case they touch only in a punctiform manner. However, in the real case, a flattening and thus a contact surface arises at the contact point 22 due to the elasticity. A characteristic pressure distribution (surface pressure) arises on the contact surface in both bodies, wherein the pressure is always highest in the middle. If, as here, a ball outer surface and a flat flange surface touch, a touch- or contact-ellipse results. Due to the comparatively high surface pressure, with roller bearings having straight flanges, a relatively poor lubricant film formation generally results at higher effective forces. In addition, in comparison to spherical flanges, straight or flat flanges lead to smaller contact ellipses between the roller end side and the flange surface facing this, for which reason an overlapping of the contact ellipse with the flange edges can result only with extreme loads. Likewise, with straight- or flat-embodied flanges, there is a low sensitivity of the contact point 22 to alignment errors, so that a defined contact point 22 between roller 13 and flange is possible. While on the one hand a greater skewing of the roller 13 is made possible with flat-embodied flanges, on the other hand a relatively poor guiding of the rollers results during operation.

Tapered roller bearings in the large bearing field can, as shown with reference to FIG. 1, be embodied with spherical flanges 23, which compared to straight or flat flanges has a lower surface pressure between the roller end side 21 and the flange surface facing towards the roller 13 as a consequence. In addition, spherically-designed flanges 23 lead, compared to straight flanges, to larger contact ellipses between the roller end side 21 and the opposing flange surface, so that overlapping of the contact ellipse with the flange edges and thus edge stresses can frequently result. In general, with spherically-designed flanges 23 there is a higher sensitivity of the contact point 22 to alignment errors than is the case with flat- or straight-embodied flanges. Although on the one hand spherical flanges have a lower skewing of the roller 13 as a consequence, on the other hand due to the narrow osculation between the roller end side 21 and the flange surface facing towards roller 13, the roller 13 can be guided better during operation. A defined contact point 22 between the roller 13 and the flange 23 is also theoretically possible with spherical flanges by a different choice of the radii of curvature (and/or their origins) of roller end surface 21 and spherical flange surface.

However, one of the main disadvantages of a spherical flange is the resulting sensitivity of the contact point 22 between the roller end side 21 and the flange 23 to alignment

error. Deviations in the raceway angle, roller angle, flange radius, as well as roller end side radius have a decisive influence thereon.

SUMMARY

Therefore, in one aspect of the present teachings, techniques are disclosed to reduce this sensitivity of the contact point between the roller end side and the flange to alignment error.

In order to reduce the sensitivity of the roller-flange contact point to alignment error, but nevertheless to be able to ensure a sufficient guiding of the roller during operation, an appropriately optimized flange surface geometry is proposed herein. For this the proposed flange surface geometry has on the one hand a spherical section which should ensure the roller guiding as well as a low Hertzian pressure. On the other hand, the proposed flange geometry also has what is referred to below as an opening section, which produces a larger gap dimension, i.e. a larger spacing, between the roller end surface and the flange surface facing towards this, than would be provided with a purely spherically formed flange.

For this, exemplary embodiments of the present invention provide a bearing ring for a roller bearing having a raceway for at least one roller, which has an at least partially spherically formed roller end surface having a first curvature. For transferring axial forces, the bearing ring has a flange disposed on an end-side of the roller, which flange has a flange surface facing towards the roller end side, wherein the flange surface comprises a spherical flange surface section having a second curvature, which can be designed as constant or variable on the spherical flange surface section. The first and the second curvature are chosen so as to achieve a first gap dimension (which depending on the roller-flange geometry can also be a mean gap dimension) between the spherically-designed roller end surface and the spherical flange surface section. In addition, the flange surface has an opening flange surface section having a lesser-than-the-second curvature for a larger-than-the-first gap dimension between the roller end surface and the opening flange surface section.

According to a further aspect, a method is also provided for manufacturing a bearing ring for a roller bearing having a roller raceway for at least one roller, wherein the roller has a spherically-designed roller end surface having a first curvature. The method comprises a step for providing a flange disposed on an end-side of the roller for transferring axial forces, which flange has a flange surface facing towards the roller end surface, wherein the provision step in turn includes the following steps:

Provision of a spherical surface section of the flange surface with a second curvature, wherein the second curvature is chosen with respect to the first curvature such that a first gap dimension is achieved between the spherically embodied roller end surface and the spherical flange surface section; and

Production of an opening surface section of the flange surface with a lesser-than-the-second curvature, so that a larger-than-the-first gap dimension results between the spherically embodied roller end surface and the opening surface section.

According to exemplary embodiments, the second curvature of the spherical flange surface section can be designed as constant or variable.

According to exemplary embodiments, a first tangential plane to a hypothetical purely spherically extending flange surface, instead of the opening surface section in this region, and a second tangential plane to the opening flange surface

section enclose a maximum angle α , which is larger than 0° and smaller than or equal to 30° .

According to exemplary embodiments, the spherical flange surface section extends over an area of at least one third of the flange surface.

According to some exemplary embodiments, the spherical flange surface section extends from a corner region between the raceway and the flange surface to a radial end of the flange or a flange edge. In a bearing inner ring, the spherical flange surface section can thus lie in a lower region of the flange, which lower region faces towards an inner race and adjacent to a flange or groove edge, wherein the opening flange section in this case, bordering the spherical flange surface section, lies in the upper region of the flange. In a bearing outer ring it is correspondingly reversed, i.e. the spherical flange section here lies in an upper region of the flange bordering a flange or groove edge, wherein the opening flange section is then located in the lower region of the flange, which lower region faces away from the outer race.

However, according to other exemplary embodiments it is also possible that the spherically-designed section does not border the flange edge between raceway and flange, so that opening flange surface regions are located above and below the spherical section. In this case, a first opening flange surface section extends from a corner region between the raceway and the flange surface, wherein the spherical flange surface section connects to the first opening flange surface section in the radial direction. On a region of the spherical flange surface section facing away from the first opening flange surface section, a second opening surface section extends in the radial direction from the spherical flange surface section to a radial end of the flange.

According to exemplary embodiments the roller bearing is a tapered roller bearing. The roller is then accordingly formed as a tapered roller.

Due to the proposed roller-flange contact geometry in the form of a spherical flange surface section in combination with a directly-bordering opening flange surface section, the roller guiding during operation can be preserved due to the spherically formed flange section. The spherical flange surface section additionally provides for a low Hertzian pressure.

The opening flange section can reduce a size of the contact ellipse in comparison to purely spherical flanges, whereby an overlapping of the contact ellipse with the flange edges can be avoided. Due to the opening flange regions, there is also a lower sensitivity of the location of the contact point between roller end side and flange to alignment errors, so that a defined contact point between roller and flange is made possible.

The proposed roller-flange geometry is designed such that in the ideal geometry the contact point or region between roller end side and flange lies in the spherical flange region.

The sensitivity of the contact point to alignment errors is comparable in the spherical flange region to the sensitivity of a purely spherical flange. However, the contact point shifts due to deviations in shape in the opening flange region; thus the sensitivity is greatly reduced, which can prevent a "wandering" of the theoretical contact point out over the flange edges and thus also can prevent the above-mentioned high edge stresses.

BRIEF DESCRIPTION OF THE DRAWINGS

Exemplary embodiments of the present invention are explained in more detail below with reference to the accompanying figures:

FIG. 1 shows a schematic longitudinal section through a roller bearing having spherical flanges;

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FIG. 2 shows a schematic longitudinal section through a roller bearing having a flange surface with a spherical flange surface section and an opening flange surface section according to an exemplary embodiment of the present invention;

FIG. 3 shows an enlarged representation of the roller-flange contact region according to FIG. 2;

FIG. 4 shows a schematic longitudinal section through a roller bearing having a flange surface with a spherical flange surface section and an opening flange surface section according to a further exemplary embodiment of the present invention;

FIG. 5 shows an enlarged representation of the roller-flange contact region according to FIG. 4; and

FIG. 6 shows a schematic flow diagram for a method of manufacture for a roller bearing having a flange having a spherical flange region and an opening flange region.

DETAILED DESCRIPTION OF THE INVENTION

FIG. 2 shows, in a schematic longitudinal section, a section of a roller bearing according to an exemplary embodiment of the present invention.

A part of a bearing ring 30 is depicted for the roller bearing having a raceway 14 for at least one roller 13. The roller 13 has a curved roller end surface 21 having a first curvature, which curved roller end surface 21 can be formed e.g. partially spherical. The bearing ring 30 has a flange 31 for transferring axial forces, which flange 31 is disposed end-side to or on the end side of the roller 13 having a flange surface 32 facing towards the roller end surface.

As depicted in the enlarged view of FIG. 3 in addition to FIG. 2, the flange surface 32 facing towards the roller end surface 21 comprises a spherical flange surface section 33 having a second curvature, wherein the first and the second curvature are chosen so as to achieve a first gap dimension d_1 , having a maximum $d_{1,max}$, between the spherically formed roller end surface 21 and the spherical flange surface section 33. The flange surface 32 also has an opening flange surface section 34 having a lesser curvature than the second curvature, so that a greater gap dimension d_2 than the (maximum) first gap dimension $d_{1,max}$ is achieved between the roller end surface 21 and the opening flange surface section 34. According to exemplary embodiments, the second curvature of the spherical flange surface section 33 can be designed as constant or variable.

In this case curvature shall be understood to mean a change of direction per unit length. The curvature e.g. of a straight line is everywhere equal to zero, since its direction does not change. A circle having a radius r has the same curvature everywhere (namely $1/r$), since its direction changes everywhere equally strongly. With all other curves the curvature changes from curve point to curve point. The curvature of a curve at a point thus indicates how strongly the curve deviates, in the immediate vicinity of the point, from a straight line. In a curved surface, such as for example the flange surface 32, one notices its curvature for example in a quadratically increasing deviation of the surface from its tangential plane. A stronger curvature makes itself noticeable as a stronger deviation from the plane. For the present case this also means, for example, that the spherical flange surface section 33 deviates more strongly per path length from a tangential plane to the opening flange surface section 34 of the flange surface 32 than regions of the opening flange surface section 34 itself.

The spherical roller end surface 21 is described by a first radius R_{21} , which can have its origin substantially on the axis of rotation 20 of the roller 13. The spherical flange surface

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section 33 is described by a second radius R_{33} , which is larger than the first radius R_{21} (i.e. $R_{21} < R_{33}$) and which can have its origin substantially on the axis of rotation 18 of the roller bearing 10. The radii R_{21} and R_{33} or the surfaces 21 and 33 described thereby can indeed be in principle disposed concentric, i.e. with an identical origin on the roller rotational axis 20. In order to at least theoretically obtain a defined contact point 22 between the roller end surface 21 and the spherical flange surface section 33, the radii R_{21} and R_{33} or the surfaces 21 and 33 described thereby can generally speaking (as here) also be disposed non-concentric, i.e. with a non-identical origin on the roller rotational axis 20 or bearing rotational axis 18. In any case, the first R_{21} and the second radius R_{33} and their origins are chosen so as to achieve a first gap dimension d_1 between the spherically-designed roller end surface 21 and the spherical flange surface section 33, which first gap dimension d_1 can assume a maximum value $d_{1,max}$. However, the opening flange surface section 34 now has according to the invention, a lesser curvature than the spherical flange surface section 33, so that between the opening flange surface section 34 and the spherical roller end surface 21 a second gap dimension d_2 arises, which, starting from the maximum value $d_{1,max}$ out to a radial end of the opening flange surface section 34 or out to one of the flange edges 36 facing away from the raceway 14 increases up to a maximum value $d_{2,max}$, i.e. $d_{1,max} \leq d_2 \leq d_{2,max}$.

In the exemplary embodiment illustrated in FIGS. 2 and 3, the spherical flange surface section 33 extends from a corner region 35 between the raceway 14 and the flange surface 32 towards the flange edge 36 facing away from the raceway 14. In a bearing inner ring the spherical flange surface section 33 can thus for example lie in a lower region of the flange 31 adjacent to a flange or groove edge 35, which lower region faces towards an inner race 14, wherein the opening flange surface section 34 lies in this case in the upper region of the flange 31. In one exemplary embodiment for a bearing outer ring it would be correspondingly reversed, i.e. the spherical flange surface section 33 would here lie in an upper-in-a-radial-direction region of the flange adjacent to a flange or groove edge, wherein the opening flange section then would lay in the lower region of the guiding flange, which lower region faces away from an outer race. Of course the spherical flange surface section would also lay here adjacent to the race.

As is shown in FIG. 3, the spherical flange surface section 33 and the opening flange surface section 34 directly border each other in a boundary region 37. That is, in the boundary region 37, for example a boundary line, the spherical flange surface section 33 and the opening flange surface section 34 merge directly into each other. The flange outer surface 32 preferably extends continuously in the boundary region 37, i.e. without corners, creases, or edges, so that it is possible to place a tangential plane to the flange surface 32 through a boundary point in the boundary region 37.

According to exemplary embodiments, the opening flange surface section 34 makes possible an opening of the flange 31 or of the flange surface 32 relative to the spherical flange surface section 33 in an angular range α from greater than 0° to 30° , preferably in a range $0^\circ \leq \alpha \leq 30^\circ$. The opening angle α of the opening flange surface 34 relative to the spherical flange surface section 33 can be obtained, for example, by determining the largest angle between a tangential plane 38 to the spherical flange surface section that is hypothetically extended into the opening flange surface section and a tangential plane 39 to the opening flange surface segment 34. For this, the largest angle between a group of tangential planes 38 to the spherical flange surface section that is hypothetically extended into the opening flange surface section, and a group

of tangential planes **39** to the opening flange surface section **34** can be determined. The spherical flange surface section that is hypothetically extended into the opening flange surface section is indicated in FIG. 3 by reference number **40**. The (largest) angle α thus results from a tilting of the tangential plane **39** to the opening flange surface section **34** towards the tangential plane **38** to the imaginary spherical flange surface section which extends into the opening flange surface section of a completely spherically-designed flange having radius R_{33} . In this respect, each of the tangential planes to the radially-outer-lying flange ends of the opening flange surface section **34** and of the corresponding flange ends of the imaginary and spherical flange surface section, which extends into the opening flange surface section, are meant.

According to exemplary embodiments of the present invention, the spherical flange surface section **33** extends over a flange surface section corresponding to at least one third of a flange height h of the flange **31**. That is, a height h_{33} of the spherical flange surface section **33** corresponds to at least one third of the entire flange height h , i.e. $h_{33} \geq \frac{1}{3}h$. Accordingly, a height h_{34} of the opening flange surface section **34** directly adjacent thereto is at most two thirds of the total flange height h , i.e. $h_{34} \leq \frac{2}{3}h$. Preferably a height h_{33} of the spherical flange surface section **33** falls within a range of $\frac{1}{3}h \leq h_{33} \leq \frac{2}{3}h$ and accordingly the height h_{34} of the opening flange surface section **34** falls within a range $\frac{1}{4}h \leq h_{34} \leq \frac{2}{3}h$.

There are different possibilities for the embodiment of the opening flange region **34**. Although FIGS. 2 and 3 illustrate an exemplary embodiment, wherein the opening flange surface section **34** is formed by a further spherical, but less strongly curved flange surface section **34** connecting radially to a spherical flange surface section **33**, and which thus is defined by a radius R_{34} larger than the second radius R_{33} , other exemplary embodiments of the present invention can also be provided, which are formed by a straight or flat flange surface section connecting tangentially to the spherical flange surface section **33**. This flat flange surface section then virtually has an infinite radius R_{34} , which is consequently also larger than the radius R_{33} describing the spherical flange surface section **33**. Further, the radius R_{34} of the opening region **34**—starting from R_{33} —can also continuously increase within the opening flange surface section **34** from inside to outside.

After exemplary embodiments have been described with reference to FIGS. 2 and 3, wherein the spherical section **33** is disposed in the lower region of the flange **31** (adjacent to the flange edge/groove edge **35**) and the opening flange section **33** is disposed in the upper region of the flange **31**, other exemplary embodiments are subsequently described with reference to FIGS. 4 and 5, wherein the spherically-designed section **33** does not border the flange edge **35**, so that an opening flange region **34** is respectively located above and below the spherical section **33**.

According to the exemplary embodiment illustrated in FIGS. 4 and 5, a first opening flange surface section **34a** extends from a corner region **35** between the raceway **14** and the flange surface **32**. Directly bordering thereto, the spherical flange surface section **33** connects in the radial direction to the first opening flange surface section **34a**. A second opening flange surface section **34b** connects directly in the radial direction to an end of the spherical flange surface section **33**, which end is facing away from the first opening flange surface section **34a**; the second opening flange surface section **34b** extends out to the radial end or the edge **36** of the flange **31**.

According to FIGS. 4 and 5, the spherical flange surface section **33** is thus embedded between two opening flange surface sections **34a, b**, which border different ends of the spherical flange surface section **33**. The height h_{33} of the

spherical flange surface section **33** here also corresponds to at least one third of the total flange height h , i.e. $h_{33} \geq \frac{1}{3}h$. In exemplary embodiments the height h_{33} of the spherical flange surface section **33** preferably falls within a range $\frac{1}{3}h \leq h_{33} < \frac{1}{2}h$. Accordingly, the sum of the heights h_{34a} , h_{34b} of the opening flange surface section **34a, b** directly bordering thereon is at most two thirds of the total flange height, i.e. $(h_{34a} + h_{34b}) \leq \frac{2}{3}h$. Preferably, the heights, h_{34a} , h_{34b} of the opening flange surface sections **34a, 34b** each do not fall below $\frac{1}{6}h$. The flange sections **34a, 34b** can also have different heights h_{34a} and h_{34b} (e.g. $h_{33} = \frac{1}{3}h$; $h_{34a} = \frac{1}{6}h$; $h_{34b} = \frac{1}{6}h$).

In all exemplary embodiments, the roller-flange contact geometry is chosen such that the contact point **22** lies in the region of the spherical flange surface section **33**. The opening regions **34**, which can also be formed spherical, however with lesser curvature than the spherical region **33**, or can be flat or straight, respectively make possible an opening α of the flange **31** that guides the roller **13**, of approximately 0° to 30° in comparison to a flange geometry with a completely spherically-designed flange.

For the sake of completeness, a manufacturing method **60** for manufacturing a bearing ring for a roller bearing having a raceway for at least one roller **13** will now be explained with reference to FIG. 6. The roller **13** has a spherically or partially spherically formed roller end surface **21** having a first curvature.

The manufacturing process **60** includes a step **61** for preparing a flange **31** disposed on an end side of the roller for transferring axial forces, which flange **31** has a flange surface **32** facing towards the roller end surface **21**.

The step **61** can in turn be subdivided in a first substep **611** for preparing/producing a spherical surface section **33** of the flange surface **32** having a second curvature, wherein the second curvature is chosen with respect to the first curvature so as to achieve a first gap dimension d_1 between the spherically-formed roller end surface **21** and the spherical flange surface section **33**.

According to a second substep **612**, an opening surface section **34** of the flange surface **32** is also provided or produced, which opening surface section **34** has a lesser-than-the-second curvature for an enlarged gap dimension d_2 , with respect to the first gap dimension d_1 , between the roller end surface **21** and the opening flange surface section **34**. It has already been discussed in detail in the foregoing.

In summary, the proposed flange shape is thus to be sought, in order to reduce the sensitivity of the roller-flange contact point **22** to alignment errors and skewing of the bearing in use, but nevertheless to be able to ensure a sufficient guiding of the roller **13** on the raceway during operation. The inventive flange geometry has a spherical section **33**, which can ensure the roller guiding as well as a low Hertzian pressure. Furthermore, the proposed flange geometry has at least one opening section **34**, which produces a larger gap dimension between the roller end surface **21** and the flange surface **32** than would be provided by a purely spherical flange.

The spherical section **33** can for example be used in the lower region of the flange **31** (bordering the flange edge/groove edge), wherein in this case the opening flange section **34** lies in the upper region of the flange (see FIGS. 2 & 3). It is also possible that the spherically embodied section **33** does not border a flange edge, so that opening flange regions **34a, b** are located above and below the spherical section (see FIGS. 4 & 5).

The inventive roller-flange-contact-geometry of a spherical flange in combination with an opening section has the following advantageous features:

The roller guiding during operation is preserved due to the spherical flange portion 33,

Low Hertzian pressure due to the spherical flange portion 33,

The opening flange sections 34 reduce the size of the contact ellipse in comparison to purely spherical flanges, whereby an overlapping of the contact ellipse with the flange edges is avoided,

Due to the opening flange regions 34 there is a lower sensitivity to the location of the contact point 22 between the roller end side 21 and the flange 31 with respect to alignment errors,

A defined contact point 22 between the roller 13 and the flange 31 is possible.

The proposed roller-flange geometry is designed such that in the ideal geometry, the roller end side and the flange lie in the spherical flange region. The sensitivity of the contact point to alignment errors is identical in the spherical flange section to the sensitivity of a purely spherical flange. If the contact point shifts due to alignment errors in the opening flange section, then the sensitivity is greatly reduced, which can prevent a "wandering" of the theoretical contact point out over the flange edge and thus can also prevent high edge pressures.

There are different possibilities for the embodiment of the opening flange section. For one, a straight flange region would be conceivable, which connects tangentially to the arc of the spherical flange region. The opening flange region could also be defined by a radius which is larger than the radius of the spherical region.

Although the present invention has been described with reference to an embodiment having tapered roller bearings and tapered rollers, exemplary embodiments are not limited to such designs. In principle, the present invention can also be applied to other rollers and roller bearings, such as e.g. to cylindrical and barrel roller bearings.

REFERENCE NUMBER LIST

- 10 Roller bearing
- 11 Bearing inner ring
- 12 Bearing outer ring
- 13 Roller
- 14 Inner roller raceway
- 15 Outer roller raceway
- 16 Inner straight line
- 17 Outer straight line
- 18 Bearing axis of rotation
- 19 Pivot point
- 20 Roller axis
- 21 Roller end side
- 22 Contact point
- 23 Spherical flange
- 24 Flange surface radius
- 30 Part of a bearing ring
- 31 Partially spherical, partially open flange
- 32 Flange surface
- 33 Spherical flange surface section
- 34 Opening flange surface section
- 35 Corner region between raceway and flange surface
- 36 Flange edge facing away from the raceway
- 37 Boundary between spherical and opening flange surface section
- 38 Tangential plane to spherical flange surface section hypothetically extended into opening flange surface section
- 39 Tangential plane to opening flange surface section
- 40 Flange surface geometry having purely spherical flange

60 Method of manufacture

61 Step for providing a flange, disposed on an end-side of the roller, for transferring axial forces, which flange has a flange surface facing towards the roller end surface

5 611 Substep for providing a spherical surface section

612 Substep for providing an opening surface section

The invention claimed is:

1. A bearing ring for a roller bearing having a raceway for at least one roller that includes an at least partially spherically formed roller end surface having a first curvature, the bearing ring comprising:

a flange disposed on an end-side of the roller for transferring axial forces, the flange having a flange surface facing towards the roller end surface, a spherical flange surface section having a second curvature being defined on the flange surface,

wherein the first and the second curvature differ such that a first gap dimension (d_1) exists between the spherically formed roller end surface and the spherical flange surface section, and

the flange surface also has an opening flange surface section having a third curvature, which is less than the second curvature, such that a second gap dimension (d_2) exists between the roller end side and the opening flange surface section, the second gap dimension (d_2) being larger than the first gap dimension (d_1).

2. The bearing ring according to claim 1, wherein an angle (α) is defined between a first plane, which is tangential to an imaginary portion of the spherical flange surface section that is extended towards the opening flange surface section, and a second plane, which is tangential to the opening flange surface section, and the angle (α) falls within a range from greater than 0° to 30° .

3. The bearing ring according to claim 1, wherein the spherical flange surface section has a height (h_{33}) that is at least one third of a total flange height (h) of the flange in a radial direction.

4. The bearing ring according to claim 1, wherein the spherical flange surface section extends from a corner region between the raceway and the flange surface.

5. The bearing ring according to claim 1, wherein:

a first opening flange surface section extends from a corner region between the raceway and the flange surface,

45 the spherical flange surface section connects to the first opening flange surface section in the radial direction, and

on a region of the spherical flange surface section facing away from the first opening flange surface section, a second opening flange surface section of the spherical flange surface section extends out to a radial end of the flange.

6. The bearing ring according to claim 1, wherein the opening flange surface section is formed by a flat flange surface section connecting tangentially to the spherical flange surface section.

7. The bearing ring according to claim 1, wherein the opening flange surface section is formed by a further spherical flange surface section connecting to the spherical flange surface section, which further spherical flange surface section has a lesser curvature than the second curvature of the spherical flange surface section.

8. The bearing ring according to claim 1, wherein the roller bearing is a tapered roller bearing and the roller is a tapered roller.

9. The bearing ring according to claim 1, wherein the bearing ring is a bearing inner ring.

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10. A roller bearing comprising:
 at least one roller having an at least partially spherical roller
 end surface with a first curvature, and
 a bearing ring having:
 a raceway rollably supporting the at least one roller, and
 a flange extending from an axial end of the raceway and
 including a flange surface that contacts the roller end
 surface, the flange surface comprising a spherical
 flange surface section with a second curvature and a
 first opening flange surface section with a third cur-
 vature that is less than second curvature,
 wherein the first curvature differs from the second curva-
 ture such that a first gap (d_1) exists between the at least
 partially spherical roller end surface and the spherical
 flange surface section, and
 a second gap (d_2) exists between at least partially spherical
 roller end side and the first opening flange surface sec-
 tion, the second gap (d_2) being larger than the first gap
 (d_1).
11. The roller bearing according to claim 10, wherein the
 spherical flange surface section directly borders the first
 opening flange surface section.
12. The roller bearing according to claim 11, wherein:
 an angle (α) is defined between a first plane, which is
 tangential to an imaginary portion of the spherical flange
 surface section that extends towards the opening flange
 surface section, and a second plane, which is tangential
 to the opening flange surface section, and $0^\circ < \alpha \leq 30^\circ$.
13. The roller bearing according to claim 12, wherein the
 spherical flange surface section has a height (h_{33}) in a radial
 direction of the roller bearing that is at least one third of a total
 flange height (h) of the flange in the radial direction.
14. The roller bearing according to claim 13, wherein the
 spherical flange surface section directly borders a corner con-
 necting the raceway and the flange.
15. The roller bearing according to claim 13, wherein:
 the first opening flange surface section directly borders a
 corner connecting the raceway and the flange,
 the spherical flange surface section directly borders the
 first opening flange surface section in the radial direc-
 tion, and

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- a second opening flange surface section is defined on the
 flange surface, directly borders the spherical flange sur-
 face section in the radial direction and extends to a
 radially outermost end of the flange.
16. The roller bearing according to claim 15, wherein the
 second opening flange surface section is formed by a flat
 flange surface section connecting tangentially to the spherical
 flange surface section.
17. The roller bearing according to claim 15, wherein the
 second opening flange surface section is formed by a further
 spherical flange surface section directly bordering the spheri-
 cal flange surface section, the further spherical flange surface
 section having a lesser curvature than the second curvature of
 the spherical flange surface section.
18. The roller bearing according to claim 13, wherein the
 roller bearing is a tapered roller bearing and the roller is a
 tapered roller.
19. The roller bearing according to claim 18, wherein the
 bearing ring is a bearing inner ring.
20. A method for manufacturing a bearing ring having a
 raceway for at least one roller, the roller having an at least
 partially spherical roller end surface with a first curvature, the
 method comprising:
 providing a flange on an end of the raceway for transferring
 axial forces, such that the flange includes a flange sur-
 face configured to face towards the roller end surface,
 wherein said flange providing step further comprises:
 providing a spherical flange section with a second curva-
 ture on the flange surface, the second curvature differing
 from the first curvature such that a first gap dimension
 (d_1) exists between the spherically formed roller end
 surface and the spherical flange surface section, and
 providing an opening surface section having a third curva-
 ture on the flange surface such that the third curvature is
 less than the second curvature and a second gap dimen-
 sion (d_2) exists between the roller end surface and the
 opening flange surface section, the second gap dimen-
 sion (d_2) being larger than the first gap dimension (d_1).

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